Perceived Safety and Quality of Life in Calgary

DATA 602 Group 2 Project Report — Fall Survey of Calgarians

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1 Introduction

1.1 Background and Motivation

Quality of life is an important measure of how residents perceive living conditions in a city. It can reflect residents' experiences with factors such as affordability, public safety, transportation, municipal services, and the overall social and economic environment. Understanding how Calgarians perceive their quality of life can therefore provide useful information about residents' experiences and priorities.

The City of Calgary regularly conducts the **Survey of Calgarians** to measure residents' perceptions of life in Calgary, satisfaction with City programs and services, attitudes toward municipal government, perceived value for taxes, and expectations for the future. The Fall Survey is also designed as a longitudinal study, allowing changes in public sentiment to be examined over time. The City states that these survey results are used to support evidence-based decision-making, performance measurement, and the identification of areas for improvement.

1.2 Problem Statement

Although an overall measure of quality of life can describe general public sentiment, it may not fully show whether residents' experiences and perceptions differ across demographic groups or change over time. Therefore, this project uses the Fall Survey of Calgarians to investigate two aspects of residents' perceptions of life in Calgary. The first analysis examines whether the proportion of Calgarians who perceive Calgary as safe differs between the 2024 and 2025 Fall Surveys. The second analysis focuses on the relationship between age group and perceived changes in Calgary's quality of life over the past three years. Together, the two analyses provide both a time-based comparison and a demographic perspective on Calgarians' perceptions of living conditions in the city.

1.3 Data Source

The data used in this project come from the publicly available **Fall Survey of Calgarians**, published by The City of Calgary. The Fall 2025 survey was conducted by Ipsos Public Affairs from July 30, 2025 to August 28, 2025. A total of 2,500 Calgarians participated in the survey through telephone interviews. The Fall 2024 survey was also conducted by Ipsos Public Affairs, from August 8, 2024 to September 3, 2024, with a sample of 2,500 Calgarians.

For this project, both the 2024 and 2025 survey data are used in Analysis 1 to compare perceived safety across the two years. The 2025 survey data are used in Analysis 2 to examine the relationship between age group and perceived change in quality of life. ## Sampling Method and Study Population The Fall Survey of Calgarians used a random telephone sampling design. For the 2025 survey, Ipsos used random digit dialing with both cellular and landline telephone numbers. Approximately 60% of the telephone numbers were cell-phone numbers and 40% were landline numbers. This approach was used to obtain a random and statistically representative sample of Calgarians.

Target sample proportions were established for demographic characteristics such as age, gender, and city ward to improve the representativeness of the sample with respect to Calgary's adult population. The City subsequently applies survey weights based on demographic information so that the final reported results more closely reflect the composition of Calgary's population. Therefore the population of interest is adult residents of the City of Calgary aged 18 years and older. The sample for the Fall 2025 survey consists of 2,500 Calgarians who completed the telephone survey.

2 Analysis 1: Perceived Safety in Calgary

2.1 Research Question

Is there a difference in the proportion of Calgarians who perceive Calgary as safe between 2024 and 2025?

Perceived safety is one aspect of residents' experience of living in Calgary. Comparing the two fall surveys can help us understand whether the proportion of residents who perceived Calgary as safe differed between the two years. A two-sample test for equality of proportions will be used to determine whether there is statistical evidence of a difference between 2024 and 2025.

2.2 Parameter of Interest

The two populations of interest are Calgary residents aged 18 and older in 2024 and 2025.

Let p_{2024} be the proportion of Calgary residents who perceived Calgary as safe in 2024, and let p_{2025} be the proportion who perceived Calgary as safe in 2025.

The parameter of interest is:

$$p_{2025} - p_{2024}$$

This represents the difference in the population proportions between 2025 and 2024. A positive difference means that the proportion of residents who perceived Calgary as safe was higher in 2025.

2.3 Hypotheses

$$H_0 : p_{2025} - p_{2024} = 0$$

$$H_A : p_{2025} - p_{2024} \neq 0$$

The null hypothesis states that there is no difference in the population proportions of Calgary residents who perceived Calgary as safe between 2025 and 2024. The alternative hypothesis states that the population proportions are different between the two years. A significance level of $\alpha = 0.05$ will be used for the hypothesis test.

2.4 Data Preparation

The original Q10C variable has six response categories:

- 1 = Very safe
- 2 = Reasonably safe
- 3 = Somewhat unsafe
- 4 = Very unsafe
- 5 = Don't know
- 6 = Refusal

For our analysis, responses 1 and 2 were combined and classified as **Safe**, while responses 3 and 4 were combined and classified as **Unsafe**. Responses 5 (Don't know) and 6 (Refusal) were treated as missing and excluded from the analysis.

After recoding, Q10C becomes a binary categorical variable with two categories, **Safe** and **Unsafe** as shown in Table 1. The same recoding method was used for both the 2024 and 2025 survey data.

Table 1: Q10C Response Counts and Safe Percentages

Year	Total	Excluded	Valid	Safe	Unsafe	Safe (%)
2024	2500	5	2495	1831	664	73.39
2025	2500	8	2492	1843	649	73.96

2.5 Data Visualization

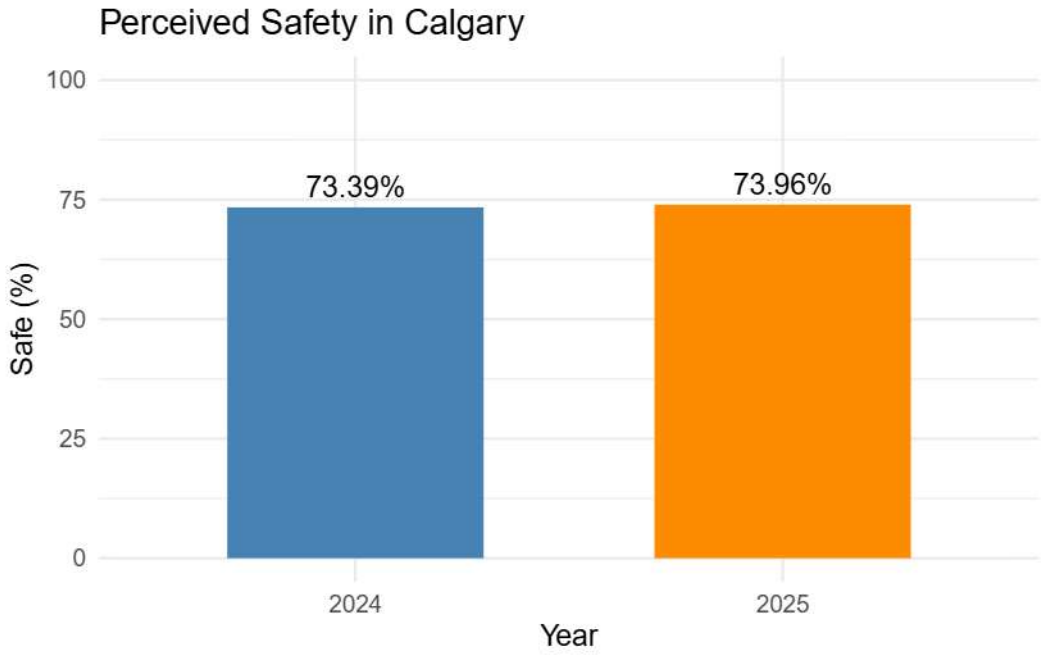


Figure 1: Percentage of respondents who perceived Calgary as safe in 2024 and 2025.

The Figure 1 shows the percentage of respondents who perceived Calgary as safe in 2024 and 2025. The percentage increased slightly from 73.39% in 2024 to 73.96% in 2025. The observed difference is approximately 0.57 percentage points, indicating that the sample proportions were very similar between the two years. However, the graph only shows the observed difference in the samples and does not tell us whether this small difference reflects a real difference between the two years. Therefore, a two-sample proportion test will be used to determine whether there is statistical evidence of a difference between 2024 and 2025.

2.6 Condition Checking

The sampling method described above supports the use of the survey samples for statistical inference. The 2024 and 2025 surveys are treated as independent samples because they were conducted as separate survey samples in the two years. The sample sizes are less than 10% of Calgary's adult population, so the independence condition is reasonable. The success-failure condition is also satisfied. In 2024, there were 1,831 Safe responses and 664 Unsafe responses. In 2025, there were 1,843 Safe responses and 649 Unsafe responses. All four counts are greater than 10. Therefore, the conditions for conducting a two-sample test for equality of proportions are satisfied.

2.7 Data Analysis

A two-sample test for equality of proportions was conducted to compare the proportion of Calgary residents who perceived Calgary as safe in 2024 and 2025.

```
    2-sample test for equality of proportions without continuity correction

data:  safety_summary$Safe[c(2, 1)] out of safety_summary$Valid[c(2, 1)]
X-squared = 0.20875, df = 1, p-value = 0.6477
alternative hypothesis: two.sided
95 percent confidence interval:
 -0.01874725  0.03014501
sample estimates:
   prop 1    prop 2
0.7395666 0.7338677
```

2.8 Inference and Interpretation

The p-value from the two-sample proportion test is 0.6477. Since the p-value is greater than the significance level of 0.05, we fail to reject the null hypothesis. There is insufficient statistical evidence to suggest that the population proportion of Calgary residents who perceived Calgary as safe was different between 2024 and 2025.

2.8.1 P-value Interpretation

Assuming that the population proportion of Calgarians who perceived Calgary as safe was the same in 2024 and 2025, the probability of observing a difference in sample proportions at least as extreme as the one observed in this study, in either direction, is approximately 0.6477.

Because this probability is large, the observed difference between the two sample proportions is reasonably consistent with random sampling variation.

3 Analysis 2: Age Group and Perceived Change in Quality of Life

3.1 Research Question

The second analysis examines whether perceptions of changes in Calgary's quality of life differ across age groups.

For this analysis, the two population variables of interest are:

- **Age Group**, a categorical variable with six categories: 18–24, 25–34, 35–44, 45–54, 55–64, and 65 and older.
- **Perceived Change in Quality of Life**, a categorical variable indicating whether respondents believe quality of life has **Improved**, **Stayed the same**, or **Worsened** over the previous three years.

Responses of "Don't know" and "Refusal" to the quality-of-life-change question are excluded because they do not represent one of the three substantive response categories being investigated. Respondents without a valid recorded age group are also excluded from this analysis.

3.2 Statistical Hypotheses

Since the Research Question is Is a Calgarian's perception of how the quality of life in Calgary has changed over the past associated with their age group? The hypothesis are

$$H_0 : \text{Age Group and Perceived Change in Quality of Life are independent.}$$
$$H_A : \text{Age Group and Perceived Change in Quality of Life are not independent.}$$

3.3 Exploratory Data Analysis

Before conducting the statistical test, the relationship between Age Group and Perceived Change in Quality of Life is explored using the observed contingency table in Table 2 and the within-age-group percentages in Table 3. A proportional bar chart in Figure 2 further illustrates the distribution of responses within each age group.

Table 2: Observed Counts for Age Group and Perceived Change in Quality of Life

	Improved	Stayed the same	Worsened
18 to 24	17	50	43
25 to 34	31	57	122
35 to 44	38	118	201
45 to 54	31	117	233
55 to 64	35	125	274
65 and older	84	352	500

Table 3: Distribution of Perceived Change in Quality of Life within Each Age Group

	Improved	Stayed the same	Worsened
18 to 24	15.454545	45.454545	39.090909
25 to 34	14.761905	27.142857	58.095238
35 to 44	10.644258	33.053221	56.302521
45 to 54	8.136483	30.708661	61.154856
55 to 64	8.064516	28.801843	63.133641
65 and older	8.974359	37.606838	53.418803

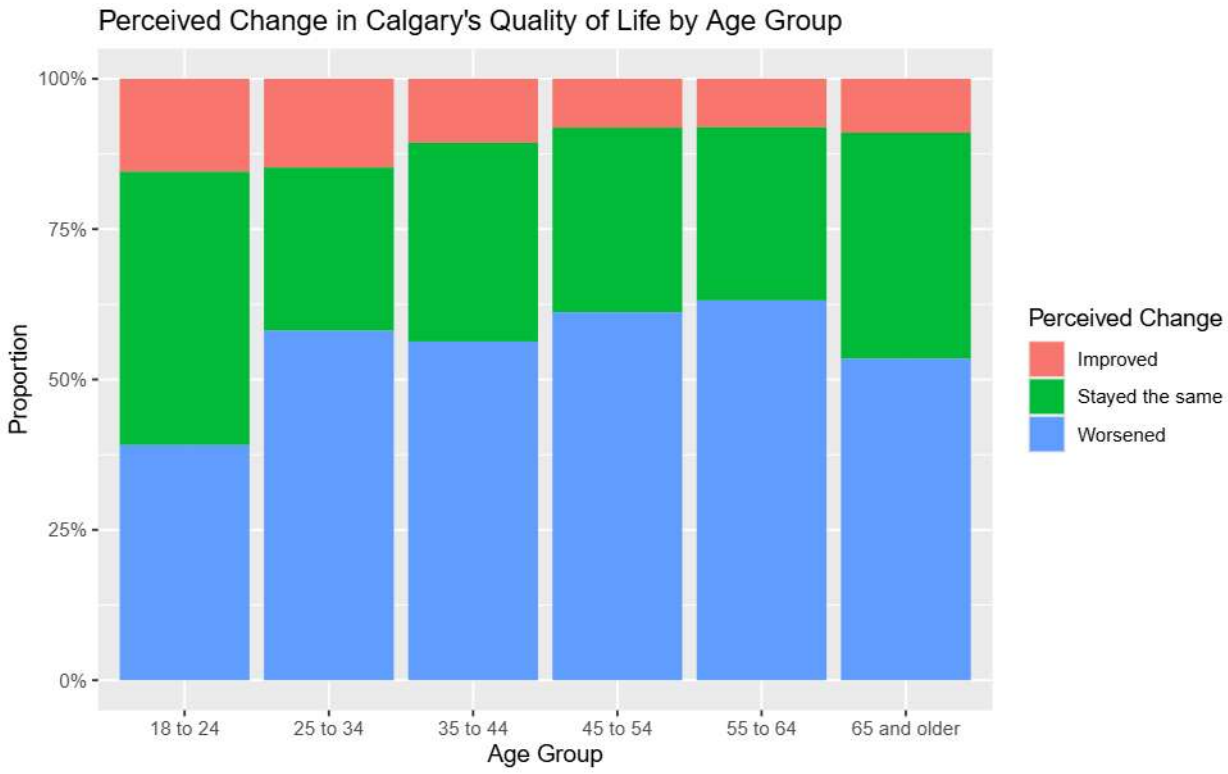


Figure 2: Distribution of Perceived Change in Calgary's Quality of Life by Age Group

The within-age-group percentages suggest that perceptions of changes in quality of life differ across age groups. In particular, respondents aged 18–24 appear less likely to report that quality of life has worsened than several of the older age groups. The graph also suggests that the distribution of perceived change in quality of life is not identical across age groups. However, these differences are descriptive only. A chi-square test of independence below is used to determine whether there is statistical evidence of an association in the population.

3.4 Compute the Expected Counts

Under the null hypothesis, Age Group and Perceived Change in Quality of Life are independent. For the chi-square test of independence, the expected cell counts are calculated using the following formula and are shown in Table 4:

$$E_{ij} = \frac{(\text{Row Total})(\text{Column Total})}{n}.$$

Table 4: Expected Counts under the Null Hypothesis of Independence

	Improved	Stayed the same	Worsened
18 to 24	10.6919	37.1046	62.2035
25 to 34	20.4119	70.8361	118.7521
35 to 44	34.7002	120.4213	201.8785
45 to 54	37.0329	128.5169	215.4502
55 to 64	42.1845	146.3946	245.4209
65 and older	90.9786	315.7265	529.2949

3.5 Compute the Value of the Test Statistic

The chi-square test of independence was carried out in R using the `chisq.test()` function. The resulting test statistic, degrees of freedom, and P-value are summarized below.

Table 5: chi-square test of independence Results

Chi.square.Statistic	Degrees.of.Freedom	P.value
40.229	10	1.54e-05

3.6 Inference and Interpretation

Since the p-value is less than the significance level of $\alpha = 0.05$, we reject the null hypothesis that Age Group and Perceived Change in Quality of Life are independent. The results provide statistically significant evidence of an association between a Calgarian's age group and their perception of how Calgary's quality of life has changed over the past three years. The observed sample distributions suggest that these perceptions vary across age groups.

3.6.1 P-value Interpretation

Assuming that Age Group and Perceived Change in Quality of Life are independent in the adult Calgary population, the probability of obtaining a chi-square test statistic at least as large as the one observed in this sample is approximately (1.54 ×10^{-5}). This very small probability provides strong evidence against the null hypothesis of independence.

4 Overall Conclusion

This project examined two aspects of Calgarians' perceptions of living conditions using the Fall Survey of Calgarians. Analysis 1 found no statistically significant difference in perceived safety between 2024 and 2025, suggesting that the proportion of Calgarians who viewed the city as safe remained relatively stable across the two survey years. In contrast, Analysis 2 found a statistically significant association between Age Group and Perceived Change in Quality of Life, indicating that perceptions of changes in quality of life varied across age groups.

Taken together, the results suggest that perceived safety in Calgary remained relatively stable from 2024 to 2025, while perceptions of changes in quality of life varied across age groups. This highlights the value of examining demographic differences in addition to overall city-wide survey results. These results could help the City of Calgary better understand how residents feel about living in the city. Instead of only looking at overall survey percentages, it may also be useful to compare results across different demographic groups. For example, future surveys or studies could look more closely at why some age groups have different views about changes in quality of life and whether factors such as affordability, transportation, public safety, or City services are related to these differences.

5 Limitations

There are several limitations to the analyses in this report. For Analysis 1, the survey measures respondents' perceptions of safety rather than actual crime or safety conditions, so the results cannot determine whether Calgary objectively became safer or less safe between 2024 and 2025. In addition, the original safety responses were combined into broader Safe and Unsafe categories, which reduces some of the detail in the original responses. Because the 2024 and 2025 surveys used separate samples, the analysis compares population proportions across years rather than changes in the same individuals over time.

For Analysis 2, the chi-square test of independence identifies an overall association between Age Group and Perceived Change in Quality of Life, but it does not show that every age group differs from every other age group, establish a causal relationship, or explain the factors behind the association. Finally, both analyses use unweighted respondent counts, while the City of Calgary applies survey weights in its official reporting. Therefore, the sample estimates in this report may differ from the City's official weighted population estimates.

6 References

City of Calgary. (2025). *Fall Survey of Calgarians 2025*. Open Calgary. https://data.calgary.ca/Help-and-Information/Fall-Survey-of-Calgarians-2025-/5wnm-smda/about_data

City of Calgary. (2024). *Fall Survey of Calgarians 2024*. Open Calgary. https://data.calgary.ca/Help-and-Information/Fall-Survey-of-Calgarians-2024-/t7u3-384g/about_data

7 Appendix: R Code

```
library(ggplot2)
library(knitr)
```

```
# Read the 2024 and 2025 survey data for analysis 1
fall2024 = read.csv("calgary_2024.csv")
fall2025 = read.csv("calgary_2025.csv")
# for analysis 2
fall2025.df =read.csv("calgary_2025.csv")
```

7.1 code for analysis 1

```
# Keep valid Q10C responses (1-4)
safety2024 = subset(fall2024, q10c %in% 1:4)
safety2025 = subset(fall2025, q10c %in% 1:4)

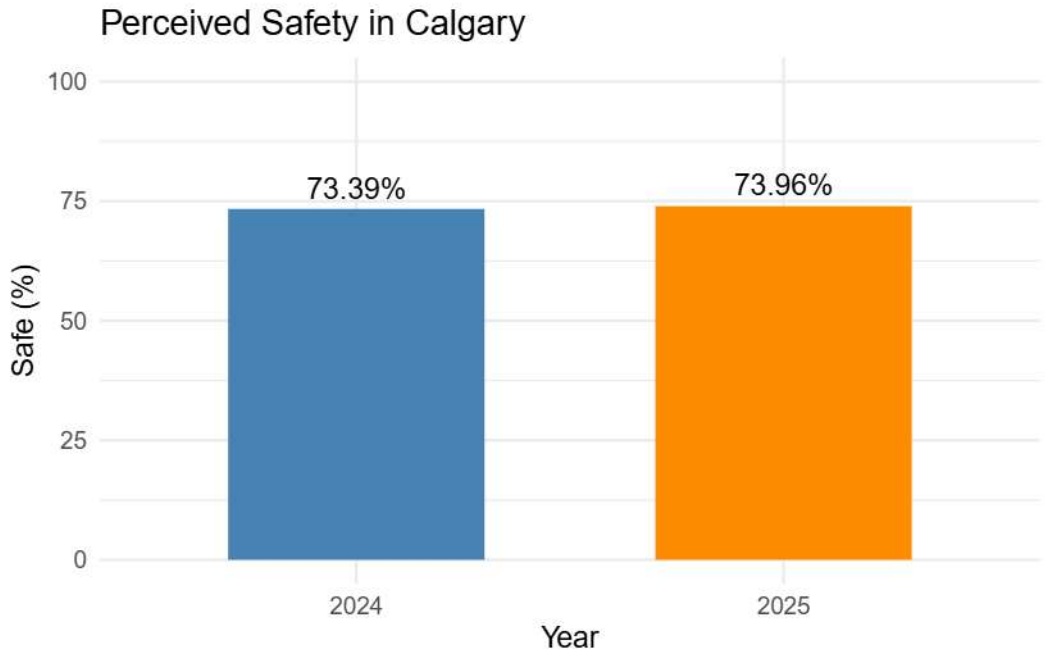
# Recode Q10C: 1 = Safe, 0 = Unsafe
safety2024$safe = ifelse(safety2024$q10c %in% c(1, 2), 1, 0)
safety2025$safe = ifelse(safety2025$q10c %in% c(1, 2), 1, 0)

# Create a summary table
safety_summary = data.frame(
  Year = c(2024, 2025),
  Total = c(nrow(fall2024), nrow(fall2025)),
  Excluded = c(sum(!fall2024$q10c %in% 1:4),
               sum(!fall2025$q10c %in% 1:4)),
  Valid = c(nrow(safety2024), nrow(safety2025)),
  Safe = c(sum(safety2024$safe), sum(safety2025$safe)),
  Unsafe = c(sum(safety2024$safe == 0), sum(safety2025$safe == 0))
)
safety_summary$Safe_percent =
  round(100 * safety_summary$Safe / safety_summary$Valid, 2)
```

```
knitr::kable(
  safety_summary,
  caption = "Q10C Response Counts and Safe Percentages",
  col.names = c("Year", "Total", "Excluded", "Valid",
                "Safe", "Unsafe", "Safe (%)"),
  digits = 2)
```

```
ggplot(safety_summary,
       aes(x = factor(Year), y = Safe_percent, fill = factor(Year))) +
```

```
  geom_col(width = 0.6) +
  geom_text(
    aes(label = paste0(Safe_percent, "%")),
    vjust = -0.5,
    size = 4
  ) +
  ylim(0, 100) +
  labs(
    title = "Perceived Safety in Calgary",
    x = "Year",
    y = "Safe (%)"
  ) +
  scale_fill_manual(values = c("2024" = "steelblue",
                               "2025" = "darkorange")) +
  theme_minimal() +
  theme(legend.position = "none")
```




```
test = prop.test(
  x = safety_summary$Safe[c(2, 1)],
  n = safety_summary$Valid[c(2, 1)],
  alternative = "two.sided",
  correct = FALSE
)
test
```

```
    2-sample test for equality of proportions without continuity correction
```

```
data:  safety_summary$Safe[c(2, 1)] out of safety_summary$Valid[c(2, 1)]
X-squared = 0.20875, df = 1, p-value = 0.6477
alternative hypothesis: two.sided
95 percent confidence interval:
 -0.01874725  0.03014501
sample estimates:
   prop 1    prop 2
0.7395666 0.7338677
```

7.2 Code for Analysis 2

```
nrow(fall2025.df)
```

```
analysis2.df =subset(fall2025.df,agerange %in% 1:6 &q3 %in% 1:3)

analysis2.df$AgeGroup = factor( analysis2.df$agerange, levels = c(1,2,3,4,5,6),
    labels = c("18 to 24","25 to 34","35 to 44","45 to 54","55 to 64","65 and older"))

analysis2.df$quanlifeChange =
  factor(analysis2.df$q3,levels = c(1,2,3),labels = c("Improved","Stayed the same","Worsened"))
```

```
quanlife_age.crosstab = table(analysis2.df$AgeGroup,analysis2.df$quanlifeChange)
```

```
chisq.test =chisq.test(quanlife_age.crosstab, correct = FALSE)
chisq.test
```

```
    Pearson's Chi-squared test

data:  quanlife_age.crosstab
X-squared = 40.229, df = 10, p-value = 1.544e-05
```

```
qolage.rowpct =100 * prop.table(quanlife_age.crosstab,margin = 1)
qolage.rowpct
```

```
                Improved Stayed the same  Worsened
  18 to 24     15.454545       45.454545 39.090909
  25 to 34     14.761905       27.142857 58.095238
  35 to 44     10.644258       33.053221 56.302521
  45 to 54      8.136483       30.708661 61.154856
  55 to 64      8.064516       28.801843 63.133641
  65 and older  8.974359       37.606838 53.418803
```

```r
round(chisq.test$expected,4)
```

```
              Improved Stayed the same Worsened
18 to 24       10.6919          37.1046  62.2035
25 to 34       20.4119          70.8361 118.7521
35 to 44       34.7002         120.4213 201.8785
45 to 54       37.0329         128.5169 215.4502
55 to 64       42.1845         146.3946 245.4209
65 and older   90.9786         315.7265 529.2949
```